

Wildfire Smoke Prediction

Model Selection Process

From baseline GAM to Transformer

Model	Improves	Still limited by
GAM (baseline)	Nonlinear structure	Manual features, weak sequence modeling
Light GBM	identify useful predictors for Nonlinear tabular learning	Limited time-series handling
LSTM + Autoencoder	Sequence learning + feature compression	Slow sequential training

Therefore, we chose Transformer: Captures temporal dependence more flexibly and supports parallel training.

Transformer Advantages and Trade-offs

Why Transformer

- **Sequence-aware** models historical PM2.5 patterns directly
- **Long-range dependence** can connect distant time steps more effectively
- **Flexible representation** captures nonlinear temporal structure
- **Parallel training** more scalable than recurrent models

Trade-offs

- **Higher compute cost** larger and more complex than GAM
- **More tuning required** architecture and optimization settings matter
- **Lower interpretability** harder to explain than smooth GAM effects
- **Overfitting risk** requires careful validation and regularization

Model Architecture and Design Choice

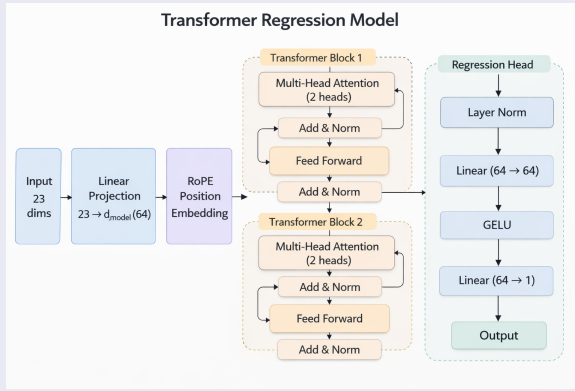
Transformer Model Structure

Architecture

- Transformer encoder
- Sequence input: **96 hours**
- $d_{\text{model}} = 64$
- 2 attention heads
- 2 Transformer layers
- dropout = 0.1

Total parameters size: 0.106M

Total parameters intentionally **kept small** to reduce overfitting.



Why Not More Complex Alternatives?

We experimented with **more complex architectures** inspired by LLaMA:

- SwiGLU feed-forward layers
- Larger hidden dimensions
- Deeper Transformer blocks

However:

- Dataset size relatively small: 26245 data entries(12.05 MB) per Zone
- Models trained **separately per zone**
- Complex variants did **not improve stability or performance**

Final decision:

Use a simpler Transformer block with strong regularization.

Training Setup

Optimizer: **AdamW**

Loss: **MSELoss**

Training parameters:

- Batch size: 256
- Epochs: 10
- Learning rate: $5e-4 \rightarrow 5e-6$ (cosine decay)
- Weight decay: 0.05

Hyperparameter search was performed using **manual exploration**.

Feature and Window Experiments

Feature set includes:

- PM2.5 lags and rolling statistics
- Fire activity indicators (FRP, fire counts)
- Temporal encoding (hour, weekday, month)

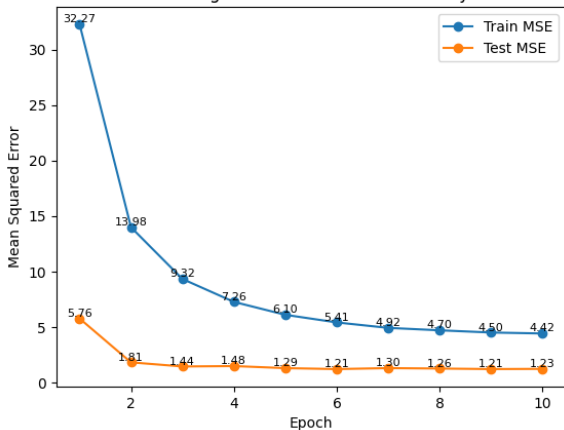
Window length experiment:

12h, 24h, 48h, 72h, **96h**, 168h

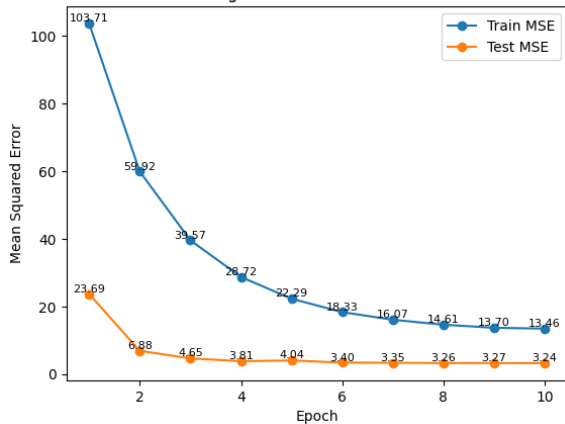
Best performance observed at **96 hours**.

Training Dynamics

Training Curve — Lower Fraser Valley



Training Curve — Southern Interior



Training monitored using **train and test loss curves** across 10 epochs.

Final Model Performance

Zone	RMSE	MAE	R ²	N_test
Lower Fraser Valley	1.108	0.659	0.843	8710
Southern Interior	1.801	1.230	0.863	8710

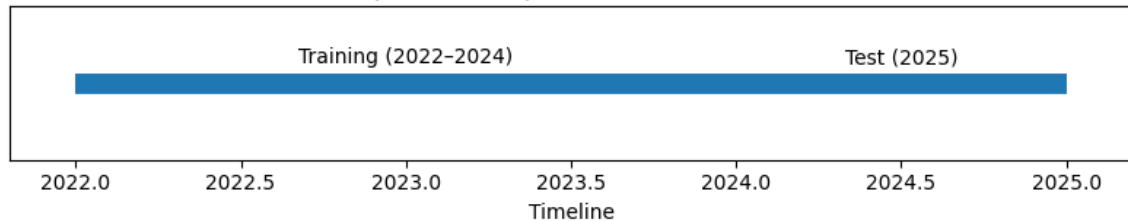
Validation Strategy and Leakage Prevention

Time-Series Evaluation

Random data splits are **inappropriate for time-series data**, because they allow future observations to influence training.

Instead we use **temporal splitting**.

Temporal Data Split for Model Evaluation



Train: **2022–2024**

Test: **2025**

Leakage Prevention

Several safeguards were applied:

- StandardScaler **fit only on training data**
- Test data transformed using training statistics
- Sequence windows constructed **after dataset split**
- No window crosses train–test boundaries

Zone-Specific Modeling

Models trained **independently** for each region:

- Lower Fraser Valley
- Southern Interior

This avoids mixing **regional wildfire patterns**.

Model predictions follow **PM2.5 trends in the unseen test year**, demonstrating generalization to new wildfire seasons.

10. Main performance comparison

Zone	model	Precision@25	Recall@25	F1@25	F2@25
Lower Fraser Valley	baseline	0.500	0.538	0.518	
	Transformer	0.538	0.538	0.538	0.538
Southern Interior	baseline	0.945	0.896	0.920	
	Transformer	0.945	0.888	0.916	0.899

- Model performance differed sharply by air zone.
- Southern Interior performed strongly: Precision 0.945, Recall 0.888, F1 0.916, F2 0.899.
- Lower Fraser Valley performed much worse: all key metrics were 0.538.
- Overall, the model appears reliable in the Southern Interior but much less dependable in the Lower Fraser Valley.

11. Diagnostics and air-zone comparison

- Diagnostics show that performance is highly region-dependent.
- The Southern Interior appears to have clearer or more stable event patterns, which the model captured well.
- The Lower Fraser Valley appears to be a more difficult forecasting environment, leading to weaker warning performance.
- This comparison shows why zone-level evaluation matters: overall results alone would hide important regional differences.

12. Limitations, uncertainty, and final takeaways

- Results should be interpreted with caution because performance is not consistent across air zones.
- The weaker Lower Fraser Valley results indicate greater uncertainty and a higher risk of unreliable warnings.
- The approach shows clear promise, especially for the Southern Interior.
- Main takeaway: the model has value, but further refinement and zone-specific tuning are needed before broader operational use.